

URITECT

Bayesian UTI Scoring System

Version 1.0

Model version	uti_bayesian_lr_candidate_v1_20260917
Target condition	Acute uncomplicated lower urinary tract infection
Fixed prior	50%
Purpose	Technical specification and ten-expert content validation
Status	Literature-derived candidate for review

Prepared for review by ten registered medical technologists. Clinical scope and referral wording should additionally be reviewed by a licensed physician.

This document is a thesis research specification. Content validation does not establish diagnostic accuracy or calibration against urine culture.

Model version: `uti_bayesian_lr_candidate_v1_20260917` Method: Grouped likelihood-ratio Bayesian screening estimate Target: Acute uncomplicated lower urinary tract infection (UTI) Fixed starting probability (prior): 50% Review panel: Ten registered medical technologists Status: Literature-derived candidate for content validation

In simple terms: Uritect starts at 50%, then adjusts that estimate using the urine-strip pattern and reported symptoms. Evidence supporting UTI raises the estimate; evidence suggesting another cause lowers it.

1. Intended Use

This module estimates how strongly the available findings support acute uncomplicated lower UTI. It is intended for nonpregnant adult women aged 18 to 64 years with acute urinary symptoms and without a catheter, known urinary tract abnormality, immunocompromise, or systemic signs suggesting complicated infection.

The result is screening support only. It is not a diagnosis, treatment or antibiotic recommendation, locally calibrated probability, proof that UTI is present or absent, or a substitute for professional assessment and urine culture when indicated.

The ten-analyte k-NN image classifier and this Bayesian module must be evaluated separately. An incorrect strip classification can affect the Bayesian result.

For reviewers: Judge whether the inputs, evidence use, wording, and safeguards are appropriate. This review does not measure diagnostic sensitivity, specificity, or calibration against urine culture.

2. Terms Used in This Document

Technical term	Simple explanation
Prior probability	Starting estimate before the current findings are considered.
Odds	A mathematical form of probability that allows evidence to be multiplied.
Likelihood ratio (LR)	A number showing how much a finding changes the odds of UTI.
LR greater than 1	Raises the estimated likelihood of UTI.
LR less than 1	Lowers the estimated likelihood of UTI.
LR equal to 1	Makes no change.
Posterior probability	Final estimate after applying the selected evidence.
Correlated findings	Findings that overlap and should not all be counted independently.
Content validation	Expert review of relevance, clarity, and appropriateness; not a clinical accuracy study.

3. Information Used by the App

3.1 Dipstick Findings

Analyte	Negative	Positive
Nitrite	Neg or Negative	Any validated non-negative category
Leukocyte esterase	Neg or Negative	Any validated non-negative category
Blood	Neg or Negative	Trace or any higher validated category

If a result is missing, invalid, or below the app's confidence requirement, it is unavailable, not negative.

3.2 Symptoms Used in the Estimate

- Dysuria or burning during urination

- Urinary urgency
- Urinary frequency
- Visible hematuria
- Vaginal discharge
- Vaginal irritation

Suprapubic or lower abdominal pain may be displayed as clinical context, but it has no numerical LR in this version.

3.3 Systemic Warning Symptoms

- Fever or chills
- Back or flank pain
- Nausea or vomiting

These warnings are not multiplied into the lower-UTI probability. They activate a separate high-priority clinical-review message because they may indicate upper UTI, pyelonephritis, systemic illness, or complicated infection.

In simple terms: The calculator estimates lower UTI. Warning symptoms follow a separate safety pathway and can override the ordinary screen message.

4. Published Evidence

These are the literature-derived LRs considered when the model was designed. The evidence is divided into short tables so it remains readable on screen and when printed.

4.1 Dipstick Findings

Finding	LR positive	LR negative
Nitrite	5.50	0.56
Leukocyte esterase	1.40	0.40
Dipstick blood	1.70	0.89

Source: Kurotschka et al. (2024), Table 4.

4.2 Urinary Symptoms

Finding	LR present	LR absent
Dysuria	1.30	0.67
Urgency	1.20	0.75
Frequency	1.10	0.71
Dysuria and urgency together	1.50	0.44
Visible hematuria	2.00	Not assigned

Sources: Kurotschka et al. (2024) for dysuria, urgency, frequency, and the combined pattern; Bent et al. (2002) for visible hematuria.

4.3 Findings Suggesting Another Cause

Finding when present	LR
Vaginal discharge	0.30
Vaginal irritation	0.20

Source: Bent et al. (2002).

These values are below 1, so their presence lowers the UTI estimate. Their absence has no assigned LR in this model.

4.4 Combined Dipstick Patterns

Pattern	LR
Nitrite positive plus leukocytes or blood positive	7.20
Nitrite, leukocytes, and blood all negative	0.22

Source: Little et al. (2006).

Important: Section 4 is the evidence library. The app does not multiply every row. The exact rules used by the app are in Section 5.

5. Exact Scoring Rules Used by the App

The app uses three evidence groups. It selects one factor only from each group, then multiplies the selected factors. This reduces double counting among related strip findings and symptoms.

5.1 Group A: Dipstick Pattern

Start at A1 and stop at the first matching rule.

Rule	Pattern	LR
A1	Nitrite positive, plus leukocytes or blood positive	7.20
A2	Nitrite positive, without either additional positive finding	5.50
A3	Nitrite not positive; blood positive	1.70
A4	Nitrite and blood not positive; leukocytes positive	1.40
A5	Nitrite, leukocytes, and blood all confirmed negative	0.22
A6	No positive rule matches and one or more results are unavailable	1.00

Plain explanation: Only the strongest matching dipstick pattern is counted. For A1, the app uses 7.20; it does not also multiply 5.50, 1.40, or 1.70.

5.2 Group B: Urinary Symptom Pattern

Start at B1 and stop at the first matching rule.

Rule	Pattern	LR
B1	Visible hematuria present	2.00
B2	Dysuria and urgency both present	1.50
B3	Dysuria present	1.30
B4	Urgency present	1.20
B5	Frequency present	1.10
B6	No supported weighted symptom is present	1.00

Plain explanation: Only one symptom factor is counted. An unchecked symptom is not assumed to be a reliably confirmed negative, so individual symptom LR- values are not used in this version.

5.3 Group C: Possible Alternate Cause

Start at C1 and stop at the first matching rule.

Rule	Pattern	LR
C1	Vaginal irritation present	0.20
C2	Vaginal discharge present	0.30
C3	Neither finding is present	1.00

Plain explanation: These symptoms can suggest a cause other than UTI, so they lower the estimate. If both are reported, 0.20 is used once.

6. Bayesian Equation

Step 1: Convert Starting Probability to Odds

```
prior probability = 0.50

prior odds = prior probability / (1 - prior probability)
            = 0.50 / 0.50
            = 1.00
```

Step 2: Apply One LR From Each Group

```
posterior odds = prior odds
x Group A dipstick LR
x Group B symptom LR
x Group C alternate-cause LR
```

Step 3: Convert Final Odds Back to Probability

```
posterior probability = posterior odds / (1 + posterior odds)
```

Because the prior odds are 1.00, the posterior odds equal the product of the three selected group factors.

In simple terms: Choose one number from A, one from B, and one from C. Multiply them. Divide the answer by one plus the answer.

7. Worked Examples

Example 1: Stronger Support for UTI

Findings: Nitrite and leukocytes positive; dysuria and urgency present; no vaginal discharge or irritation.

```
Group A = 7.20
Group B = 1.50
Group C = 1.00

posterior odds = 1.00 x 7.20 x 1.50 x 1.00 = 10.80
posterior probability = 10.80 / (1 + 10.80) = 91.5%
```

Example 2: Mixed Evidence

Findings: Nitrite positive; dysuria present; vaginal irritation present.

```
Group A = 5.50
Group B = 1.30
Group C = 0.20

posterior odds = 1.00 x 5.50 x 1.30 x 0.20 = 1.43
posterior probability = 1.43 / (1 + 1.43) = 58.9%
```

Example 3: Less Support for UTI

Findings: Nitrite, leukocytes, and blood all confirmed negative; no weighted symptom or alternate-cause symptom recorded.

```
Group A = 0.22
Group B = 1.00
Group C = 1.00

posterior odds = 1.00 x 0.22 x 1.00 x 1.00 = 0.22
posterior probability = 0.22 / (1 + 0.22) = 18.0%
```

Example 4: Insufficient Evidence

If all three relevant dipstick results are unavailable and no supported weighted symptom is recorded, the internal prior remains 50%. The app must not present that starting value as a patient-specific result. It displays Insufficient evidence instead.

8. Display Bands

Result	App label	Plain meaning
Below 20%	Lower	Less support for lower UTI, but UTI is not ruled out.
20% to below 80%	Intermediate	Evidence is mixed; clinical review is needed.
80% or above	Higher	Stronger support for lower UTI; this is not a diagnosis.
No usable evidence	Insufficient	No patient-specific estimate can be calculated.

The 20% and 80% boundaries are provisional display bands created for the Uritect interface. They are not published treatment or antibiotic thresholds. Review them separately from the published LR values.

9. Reviewer Instructions

9.1 Reviewer Information

- Reviewer name or code: _____
- Professional license number, if permitted: _____
- Position and institution: _____
- Years of urinalysis experience: _____
- Urine reagent-strip experience: _____
- Laboratory quality-assurance experience: _____
- Date reviewed: _____
- Version reviewed: URITECT Bayesian UTI Scoring System v1.0

9.2 Rating Scale

Rating	Meaning
1	Not relevant or acceptable; major revision required
2	Somewhat relevant; substantial revision required
3	Relevant and acceptable with minor revision
4	Highly relevant, clear, and acceptable as written

Ratings of 3 or 4 count as expert agreement for the Content Validity Index (CVI). Add a comment whenever the rating is 1 or 2.

10. Content Validation Checklist

Write one rating from 1 to 4 on each line.

A. Scope and Intended Use

1. Target condition is clearly limited to acute uncomplicated lower UTI. Rating: ____
2. Intended population and exclusions are appropriate and clear. Rating: ____
3. Limitations clearly identify screening support, not diagnosis. Rating: ____

Comments for Items 1-3:

B. Dipstick Evidence

4. Nitrite is an appropriate UTI-screening input. Rating: ____
5. Leukocyte esterase is an appropriate UTI-screening input. Rating: ____
6. Blood is appropriately supportive rather than diagnostic. Rating: ____
7. Missing, invalid, or low-confidence results are not treated as negative. Rating: ____
8. Individual dipstick LR values and cited basis are acceptable. Rating: ____
9. Combined-positive LR 7.20 is appropriate for Rule A1. Rating: ____
10. All-three-negative LR 0.22 is appropriate for Rule A5. Rating: ____

Comments for Items 4-10:

C. Symptom Evidence

11. Selected urinary symptoms are relevant to lower-UTI screening. Rating: ____
12. Vaginal findings are appropriately treated as alternate-cause evidence. Rating: ____
13. Symptom LR values and cited basis are acceptable. Rating: ____
14. One factor per group appropriately reduces double counting. Rating: ____

Comments for Items 11-14:

D. Mathematics and Output

15. The literature-based 50% prior is clearly explained. Rating: ____
16. The Bayesian equations are mathematically correct. Rating: ____
17. Worked examples are correct and reproducible. Rating: ____
18. Output labels are understandable. Rating: ____
19. The 20% and 80% bands are clearly identified as provisional. Rating: ____

Comments for Items 15-19:

E. Safety and Overall Acceptability

20. Systemic warnings are appropriately separated from the estimate. Rating: ____
21. Wording avoids unsupported diagnosis and treatment claims. Rating: ____
22. Module is feasible for trained health workers in an RHU. Rating: ____
23. Limitations and safety messages are complete and understandable. Rating: ____
24. Specification is acceptable for thesis research use with stated limits. Rating: ____

Comments for Items 20-24:

Overall Decision

- ☐ Accept without revision
- ☐ Accept after minor revision
- ☐ Major revision and repeat review required
- ☐ Reject for the stated intended use

Reviewer signature: _____ Date: _____

11. Summarizing the Ten Reviews

For each item:

$I\text{-CVI} = \text{number of reviewers giving rating 3 or 4} / 10$

Example with 8 agreeing reviewers:

$I\text{-CVI} = 8 / 10 = 0.80$

For the whole checklist:

$S\text{-CVI/Ave} = \text{sum of the 24 item I-CVI values} / 24$

Result	Planned action
I-CVI 0.80 or above	Retain; apply minor wording revision if needed
I-CVI 0.70	Revise and return for expert review
I-CVI below 0.70	Replace or remove unless a rationale supports repeat review
S-CVI/Ave 0.90 or above	Target for strong overall content agreement

Reviewer comments must still be examined when the target is met. A safety objection must not be dismissed by averaging scores.

Results Worksheet

Item	Reviewers rating 3 or 4	I-CVI	Decision
1			
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S-CVI/Ave: _____

Reporting limitation: CVI shows expert agreement about content and presentation. It is not diagnostic sensitivity, specificity, predictive value, calibration, or validation against culture.

12. References

1. Kurotschka PK, Gagyor I, Ebell MH. Acute Uncomplicated UTIs in Adults: Rapid Evidence Review. American Family Physician. 2024;109(2):167-174. <https://www.aafp.org/pubs/afp/issues/2024/0200/acute-uncomplicated-utis-adults.html>
2. Bent S, Nallamothu BK, Simel DL, Fihn SD, Saint S. Does This Woman Have an Acute Uncomplicated Urinary Tract Infection? JAMA. 2002;287(20):2701-2710. <https://pubmed.ncbi.nlm.nih.gov/12020306/>
3. Little P, Turner S, Rumsby K, et al. Developing clinical rules to predict urinary tract infection in primary care settings. Br J Gen Pract. 2006;56(529):606-612. <https://pmc.ncbi.nlm.nih.gov/articles/PMC1874525/>
4. Giesen LGM, Cousins G, Dimitrov BD, van de Laar FA, Fahey T. Predicting acute uncomplicated urinary tract infection in women: a systematic review. BMC Fam Pract. 2010;11:78. <https://pmc.ncbi.nlm.nih.gov/articles/PMC2987910/>
5. Lynn MR. Determination and quantification of content validity. Nurs Res. 1986;35(6):382-385. <https://pubmed.ncbi.nlm.nih.gov/3640358/>
6. Polit DF, Beck CT, Owen SV. Is the CVI an acceptable indicator of content validity? Res Nurs Health. 2007;30(4):459-467. <https://pubmed.ncbi.nlm.nih.gov/17654487/>